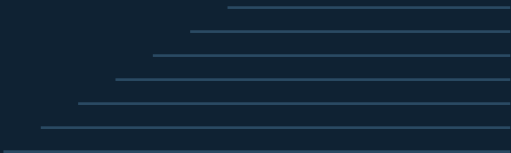


Melting Ice Trackers

40-day implementation plan

Glacier monitoring and forecasting system: architecture, phases, schedule and team split



Key changes before coding starts

The plan below is a strong basis for the project. Four adjustments matter most before the team writes code: the scope of the first prototype, the training data, the forecast target, and what “real-time” means.

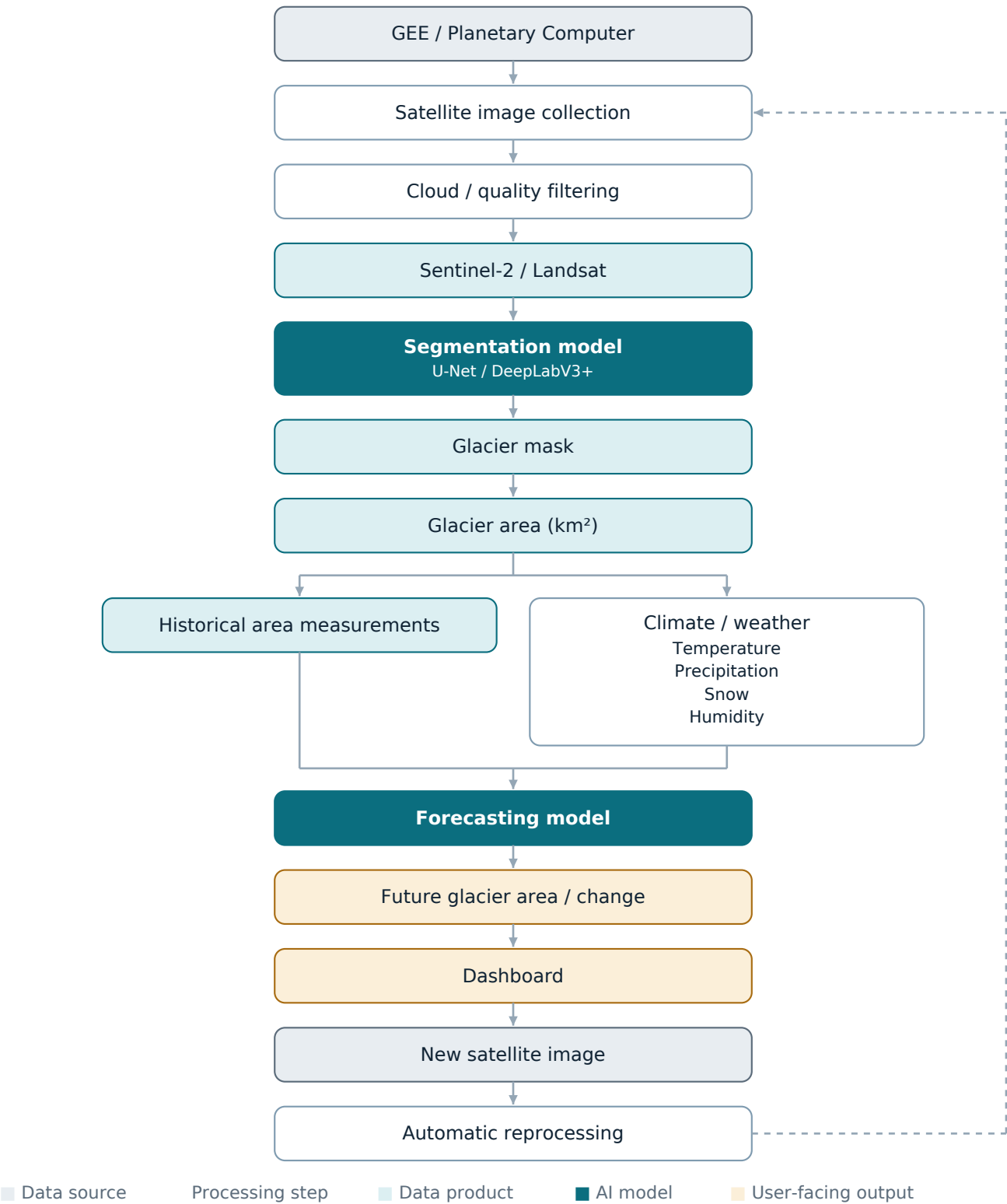
1. Start with one glacier Prove the complete pipeline on Gangotri first, then expand to two or three glaciers.	2. Build the training masks U-Net needs image + correct mask pairs. Preparing reliable masks is a core task, not a side task.
3. Define the forecast target Predict future ice-covered area change, and present it as an expected change, not as melted ice.	4. Near-real-time, not real-time Process each new satellite observation as it arrives. A watched folder makes this demonstrable.

Contents

1	Recommended architecture	6	Phase 4: Near-real-time processing
2	Start with one glacier	7	Dashboard design
3	Phase 1: Data ingestion	8	40-day schedule
4	Phase 2: Segmentation and training data	9	Team split
5	Phase 3: Forecasting	10	What to do now

1. Recommended architecture

The complete flow from satellite data to dashboard, repeating for every new observation.



2. Start with one glacier

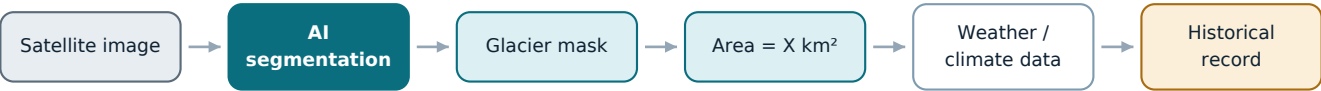
One glacier, complete pipeline, then expand.

Do not start with two or three glaciers immediately. Each new glacier brings its own geographic and data problems, and the team would spend most of the 40 days solving those instead of proving the system. Build the complete pipeline for **Gangotri Glacier** first.

Prototype observation years

2018	2019	2020	2021	2022	2023	2024	2025	2026
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Steps for each usable observation



Expansion once the prototype works



3. Phase 1: Data ingestion

Use Google Earth Engine first: satellite and environmental datasets in one ecosystem.

Four categories of data are needed:

Data	Purpose
Glacier boundary	Defines exactly which glacier is being studied
Sentinel-2 / Landsat	Satellite observations
ERA5 / ERA5-Land	Temperature, precipitation, snow and other climate variables
Historical dates	Creates the time series

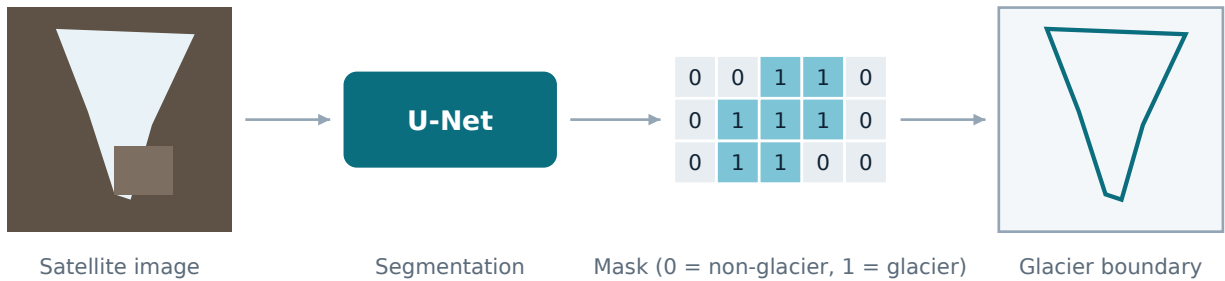
You don't need to find a dataset called “Gangotri Glacier Images”. Instead, the program asks Earth Engine for the observations it needs, which scales to any glacier.



4. Phase 2: Segmentation

Start with U-Net. Add DeepLabV3+, ResNet-50 or EfficientNet-B4 only once U-Net works.

The first goal is a model that turns a satellite image into a mask, where every pixel is labelled 0 (non-glacier) or 1 (glacier target). The mask then gives the glacier boundary.



From mask to area

Glacier area = number of glacier pixels × area represented by each pixel

Pixel area at 10 m resolution: 10 m × 10 m = **100 m²**

Example: 50,000 glacier pixels × 100 m² = 5,000,000 m² = **5 km²**

This measurement is the foundation that the entire forecasting system depends on.

The training dataset

Satellite images alone cannot train U-Net. Each training example needs an image paired with a correct mask that tells the model which pixels are glacier and which are not.

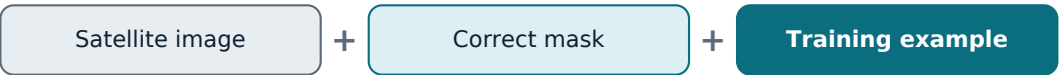


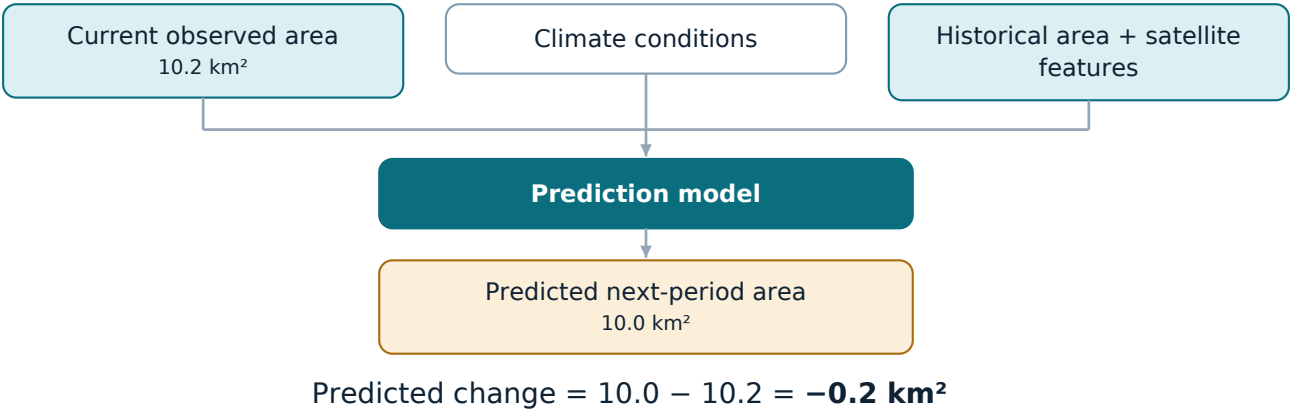
Image	Mask
Image_001.tif	Mask_001.tif
Image_002.tif	Mask_002.tif
Image_003.tif	Mask_003.tif

Creating or obtaining reliable masks is one of the most important parts of Phase 2. Glacier inventories such as GLIMS and RGI provide outlines, but the team must check that each label fits the specific satellite date and the target definition.

5. Phase 3: Forecasting

Define exactly what is predicted before choosing a model.

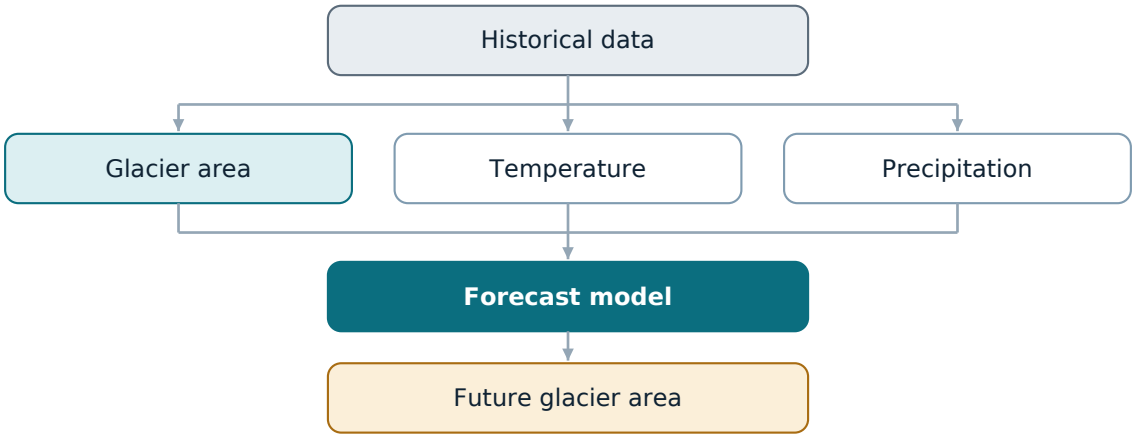
Don't start with LSTM, Autoformer and PatchTST all at once. The recommended target is **future glacier ice-covered area change**.



Display this as **“Expected change: −0.2 km²”**, not as “0.2 km² of ice will melt”. An area change is not a volume of melted ice, and the distinction is scientifically important.

Starting feature set

Previous glacier area	Previous area change	Temperature	Precipitation
Snow-related variable	Humidity / moisture	Observation quality	Season / month



Model progression

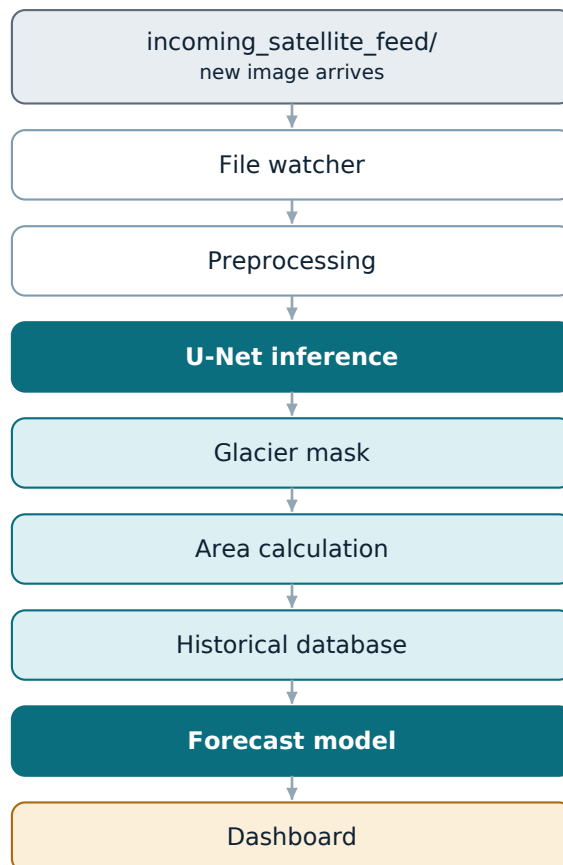
Baseline Linear Regression / XGBoost A scientifically useful reference every advanced model must beat.	Advanced LSTM Learns patterns across the sequence of observations.	Experimental PatchTST / Autoformer Only if there are enough historical observations to justify it.
--	--	--

Don't use a transformer just because it sounds more advanced. It needs enough historical observations to justify it.

6. Phase 4: Near-real-time processing

A watched folder gives a convincing live demonstration.

Sentinel-2 does not provide second-by-second imagery, so the system processes each new observation as it arrives. For the hackathon, a folder called `incoming_satellite_feed/` acts as the satellite feed: placing `new_satellite_image.tif` into it triggers the full pipeline.



What the audience sees during the demo

- 1 New satellite observation detected
- 2 Processing...
- 3 Glacier segmentation complete
- 4 Area calculated
- 5 Forecast updated

7. Dashboard design

Six pages, from today's status to processing a new observation.

Page 1: Overview

Glacier	Gangotri
Current area	10.24 km ²
Change from previous	-0.13 km ²
Predicted next period	10.08 km ²

Page 3: Historical trend

km²

time

Glacier area over time (illustrative)

Page 5: Prediction

History + climate

ML model

Future change

Page 2: Satellite analysis

Original image

AI mask

Detected boundary

Shown on an interactive map.

Page 4: Environmental factors

Temperature

Precipitation

Snow

Humidity

Shown alongside glacier-area changes.

Page 6: New observation

Upload or drop new_satellite_image.tif

Processing → Segmentation → Area → Prediction → Dashboard update

7

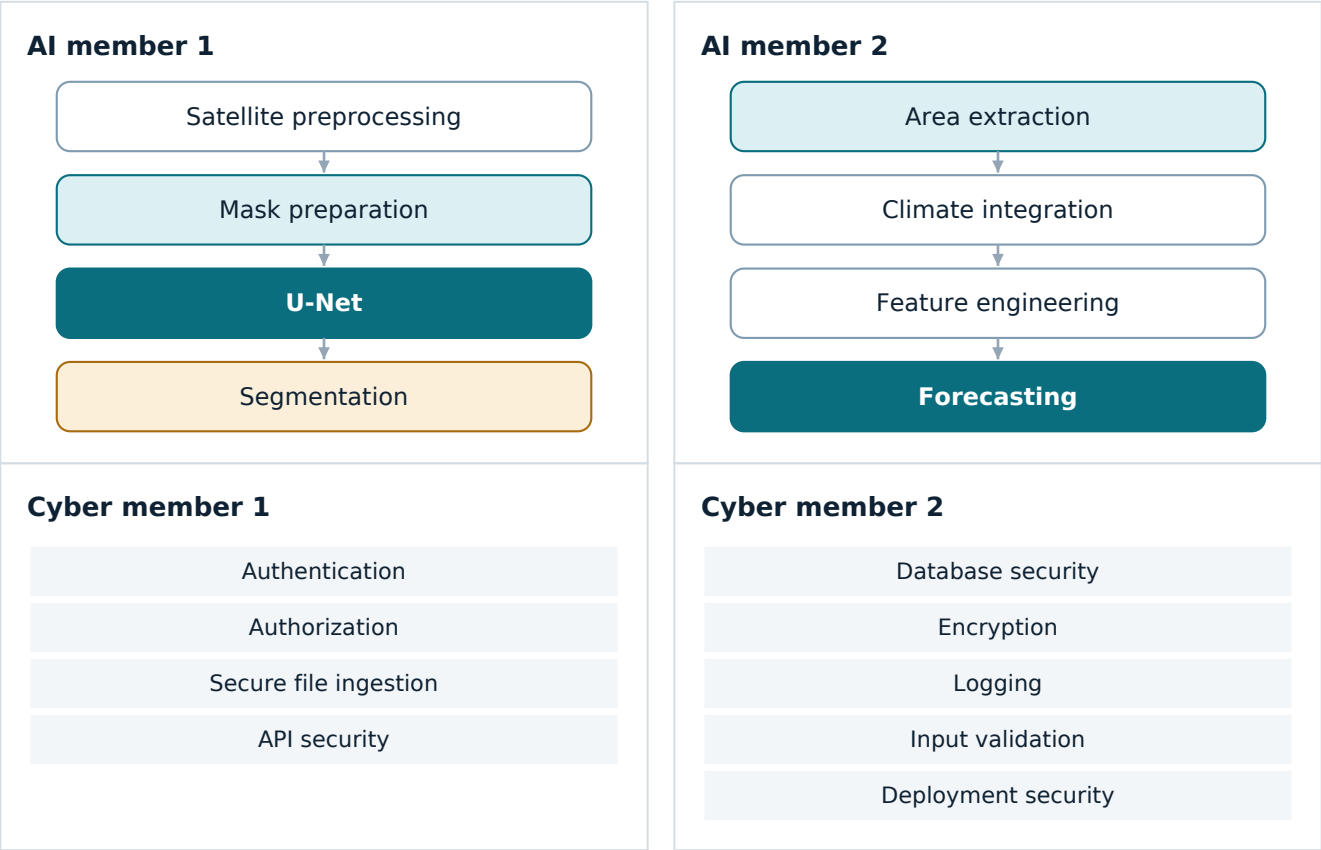
8. 40-day schedule

Each block ends with a concrete deliverable.

Days	Team focus	Deliverable	Timeline (day 1-40)
1-5	Glacier selection + GEE setup	Working data access	
6-10	Satellite + climate collection	Historical dataset	
11-14	Mask preparation	Image-mask dataset	
15-18	U-Net	Segmentation model	
19-20	Validation	IoU / Dice results	
21-23	Geospatial processing	Glacier area calculation	
24-26	Feature engineering	Forecast dataset	
27-29	Forecast model	Future-area prediction	
30	Model evaluation	Baseline vs advanced model	
31-34	Dashboard	Monitoring UI	
35-37	Automatic ingestion	New-image processing	
38-39	Integration	Complete system	
40	Testing + presentation	Final demo	

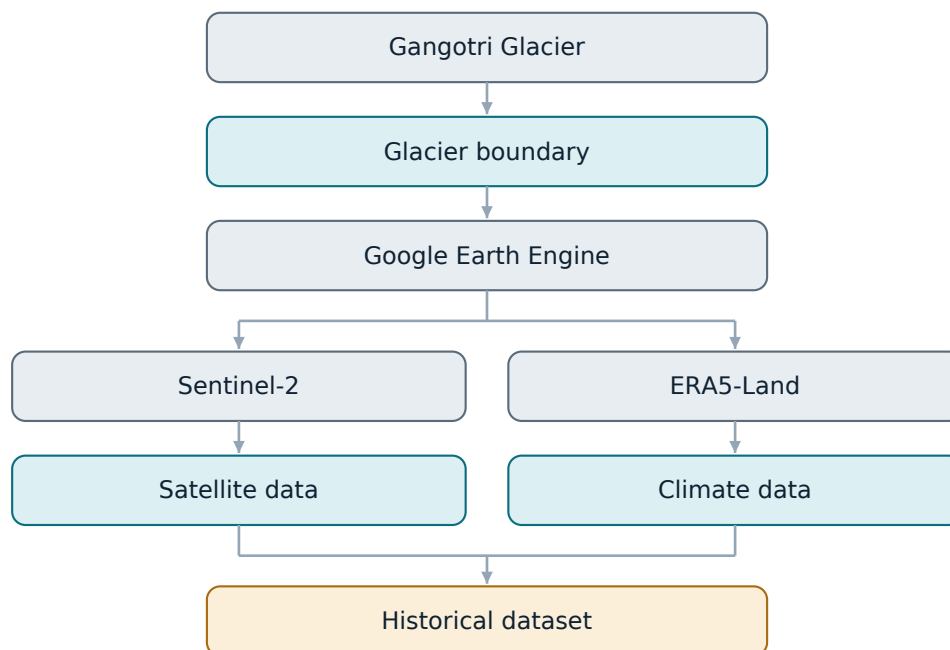
9. Team split

Two AI members build the pipeline; two cyber members secure it.



10. What to do now

Don't start training U-Net yet. First prove the data path.



Immediate objective: obtain one glacier boundary and successfully retrieve a few Sentinel-2 observations for it through Google Earth Engine. Once that works, the rest of the project becomes a sequence of engineering steps rather than a search for random datasets.